

MixBuy: Unlinkable Contingent Payment: Implementation of the Attestation Puzzle via Verifiable Witness Encryption

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1 Introduction and Motivation

Contingent payments represent a fundamental challenge in blockchain systems: how can two mutually distrustful parties—a buyer and a seller—exchange digital goods for cryptocurrency without trusted intermediaries? The problem becomes more complex when we require *unlinkability*, meaning that external observers (including payment intermediaries) cannot determine which buyer paid which seller.

The **MixBuy** protocol addresses this challenge through a novel four-party cryptographic construction called *Oracle-based Unlinkable Contingent Payment* (O-UCP). This protocol enables buyers to purchase digital products through an untrusted mixer while maintaining payment unlinkability, with security guaranteed by a semi-trusted notary who attests blockchain transactions.

2 Core Cryptographic Primitives

The MixBuy protocol relies on three fundamental cryptographic building blocks:

Adaptor Signatures: Adaptor signatures extend standard digital signatures by introducing a *pre-signature* $\tilde{\sigma}$ that becomes valid only when combined with a witness w satisfying a hard relation R . For the discrete logarithm relation $R = \{(Y, y) : Y = g^y\}$, the Schnorr-based construction allows creating pre-signatures that can be “adapted” into full signatures using the secret witness.

Witness Encryption Based on Signatures (WES): WES enables encrypting a plaintext such that decryption requires possessing a valid signature on a specified message. The construction leverages the structural similarity between BLS signatures $\sigma = H_0(m)^{sk}$ and Boneh-Franklin IBE private keys, using bilinear pairings over the BN128 curve.

Cut-and-Choose Proof System: To achieve *verifiability*—ensuring that encrypted witnesses are correctly formed without revealing them—the protocol employs a cut-and-choose technique with $\gamma = 128$ parallel instances. The Fiat-Shamir transform renders this non-interactive, achieving soundness error below 2^{-40} .

3 Verifiable Witness Encryption for a Relation (VWER)

The combination of these primitives yields **VWER**, the central construction enabling the attestation puzzle. Alice encrypts witness y (where $Y = g^y$) under the oracle’s public key and outcome message via EncR ; Bob verifies correctness via VfEncR *before* the oracle attestation exists; and given the oracle’s BLS signature, Bob recovers y via DecR . Security includes *one-wayness* (preventing extraction without attestation) and *verifiability* (guaranteeing correct decryption).

4 Implementation and Security

Our Python implementation using `py_ecc` over BN128 achieves encryption in ~ 18 seconds (dominated by 256 pairing operations), verification in ~ 12 seconds, and decryption in $\sim 5\text{--}7$ seconds. Optimizations including pairing caching could reduce time by over 50%.

The protocol provides four security properties: **Mixer Security** (mixer pays seller $\Rightarrow$ mixer receives payment), **Seller Security** (seller delivers $\Rightarrow$ seller is paid), **Buyer Security** (buyer pays $\Rightarrow$ buyer gets product), and **Unlinkability** (mixer cannot link buyers to sellers).

5 Conclusion

The attestation puzzle demonstrates that cryptographic witness encryption has matured from theoretical construct to practical tool, enabling conditional secret release based on oracle attestations without smart contracts.

References

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